

2520030583

Join and Transactions

```
Microsoft Windows [Version 10.0.26200.8973]
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C:\Users\DELL>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
'my' is not recognized as an internal or external command,
operable program or batch file.

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input
statement.

mysql> show databases;
```

```
Database
bookflow_week2
information_schema
mysql
performance_schema
sys
5 rows in set (0.00 sec)

mysql> create jt_ptl;
ERROR 1064 (42000): You have an error in your SQL syntax; check the
manual that corresponds to your MySQL server version for the right
syntax to use near 'jt_ptl' at line 1
mysql> create jt_ptl
-> ^C
mysql> create database jt_ptl;
Query OK, 1 row affected (0.09 sec)

mysql> use jt_ptl;
Database changed
mysql> create table roles(role_int primary key,role_name varchar
(50));
ERROR 1064 (42000): You have an error in your SQL syntax; check the
manual that corresponds to your MySQL server version for the right
syntax to use near 'primary key,role_name varchar(50))' at line 1
mysql> CREATE TABLE roles(role_id int primary key,role_name VARC
HAR(50));
Query OK, 0 rows affected (0.16 sec)
```

```
mysql> CREATE DATABASE jt_ptl;
Query OK, 1 row affected (0.01 sec)

mysql> USE jt_ptl;
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| jt_ptl     |
+-----+
1 row in set (0.00 sec)

mysql> CREATE TABLE roles(role_id INT PRIMARY KEY,role_name VARC
HAR(50));
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_jt_ptl |
+-----+
| roles             |
+-----+
1 row in set (0.00 sec)

mysql> DESC roles;
```

```
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| role_id | int | NO | PRI | NULL | |
| role_name | varchar(50) | YES | | NULL | |
+-----+
2 rows in set (0.00 sec)

mysql> CREATE TABLE users(user_id INT PRIMARY KEY,name VARCHAR(5
0),email VARCHAR(50),role_id INT);
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_jt_ptl |
+-----+
| roles             |
| users             |
+-----+
2 rows in set (0.00 sec)

mysql> CREATE TABLE departments(dept_id INT PRIMARY KEY,dept_nam
e VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE employees(emp_id INT PRIMARY KEY,emp_name VA
RCHAR(50),dept_id INT,manager_id INT,salary DECIMAL(10,2));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_jt_ptl |
+-----+
| departments       |
| employees         |
| roles             |
| users             |
+-----+
4 rows in set (0.01 sec)

mysql> CREATE TABLE customers(customer_id INT PRIMARY KEY,custom
er_name VARCHAR(50),city VARCHAR(50));
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_jt_ptl |
+-----+
| customers         |
| departments       |
| employees         |
| roles             |
| users             |
+-----+
5 rows in set (0.00 sec)

mysql> CREATE TABLE orders(order_id INT PRIMARY KEY,customer_id
INT,order_date DATE,total_amount DECIMAL(10,2));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_jt_ptl |
+-----+
| customers         |
| departments       |
| employees         |
| orders           |
| roles             |
| users             |
+-----+
6 rows in set (0.00 sec)

mysql> CREATE TABLE accounts(account_no INT PRIMARY KEY,holder_n
ame VARCHAR(50),balance DECIMAL(10,2));
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_jt_ptl |
+-----+
| accounts          |
| customers         |
| departments       |
| employees         |
| orders           |
| roles             |
| users             |
+-----+
7 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE students(student_id INT PRIMARY KEY,student_name VARCHAR(50));
Query OK, 0 rows affected (0.06 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_jt_pt1 |
+-----+
| accounts          |
| customers         |
| departments       |
| employees         |
| orders            |
| roles             |
| students          |
| users             |
+-----+
8 rows in set (0.00 sec)

mysql> CREATE TABLE faculty(faculty_id INT PRIMARY KEY,faculty_name VARCHAR(50));
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_jt_pt1 |
+-----+
| accounts          |
| customers         |
| departments       |
| employees         |
```

```
employees
|
| faculty
|
| orders
|
| roles
|
| students
|
| users
+-----+
9 rows in set (0.00 sec)

mysql> CREATE TABLE courses(course_id INT PRIMARY KEY,course_name VARCHAR(50),faculty_id INT);
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_jt_pt1 |
+-----+
| accounts          |
| courses           |
| customers         |
| departments       |
| employees         |
| faculty           |
| orders            |
| roles             |
| students          |
| users             |
+-----+
10 rows in set (0.00 sec)

mysql> CREATE TABLE enrollments(enrollment_id INT PRIMARY KEY,student_id INT,course_id INT);
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_jt_pt1 |
+-----+
| accounts          |
| courses           |
| customers         |
| departments       |
| employees         |
| enrollments       |
| faculty           |
| orders            |
| roles             |
| students          |
| users             |
+-----+
11 rows in set (0.00 sec)

mysql> INSERT INTO roles VALUES(1,'Admin'),(2,'Manager'),(3,'Employee');
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO users VALUES(101,'Rahul','rahul@gmail.com',1),(102,'Priya','priya@gmail.com',2),(103,'Arjun','arjun@gmail.com',3),(104,'Sneha','sneha@gmail.com',NULL);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> desc employees;
+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+
| emp_id     | int           | NO   | PRI | NULL    |       |
| emp_name   | varchar(50)   | YES  |     | NULL    |       |
| dept_id    | int           | YES  |     | NULL    |       |
| manager_id | int           | YES  |     | NULL    |       |
| salary     | decimal(10,2) | YES  |     | NULL    |       |
+-----+
5 rows in set (0.01 sec)

mysql> SELECT * FROM roles;
+-----+
| role_id | role_name |
+-----+
| 1       | Admin    |
| 2       | Manager  |
| 3       | Employee |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM users;
+-----+
| user_id | name   | email          | role_id |
+-----+
| 101     | Rahul | rahul@gmail.com | 1       |
| 102     | Priya | priya@gmail.com | 2       |
| 103     | Arjun | arjun@gmail.com | 3       |
| 104     | Sneha | sneha@gmail.com | NULL    |
+-----+
```

```
mysql> INSERT INTO departments VALUES(1,'CSE'),(2,'ECE'),(3,'EEE');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM departments;
+-----+
| dept_id | dept_name |
+-----+
| 1       | CSE      |
| 2       | ECE      |
| 3       | EEE      |
+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO employees VALUES(1,'Amit',1,NULL,80000),(2,'Ravi',1,1,60000),(3,'Neha',2,1,45000),(4,'Kiran',3,2,55000);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM employees;
+-----+
| emp_id | emp_name | dept_id | manager_id | salary |
+-----+
| 1       | Amit     | 1       | NULL       | 80000.00 |
| 2       | Ravi     | 1       | 1          | 60000.00 |
| 3       | Neha     | 2       | 1          | 45000.00 |
| 4       | Kiran    | 3       | 2          | 55000.00 |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO customers VALUES(1,'Ramesh','Hyderabad'),(2,'Sita','Vijayawada'),(3,'Arun','Vizag');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM customers;
+-----+
| customer_id | customer_name | city      |
+-----+
| 1           | Ramesh        | Hyderabad |
| 2           | Sita          | Vijayawada |
| 3           | Arun          | Vizag     |
+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO orders VALUES(201,1,'2026-08-01',2500),(202,2,'2026-08-02',3500),(203,1,'2026-08-03',1500),(204,3,'2026-08-04',5000);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM orders;
+-----+
| order_id | customer_id | order_date | total_amount |
+-----+
| 201      | 1           | 2026-08-01 | 2500.00      |
| 202      | 2           | 2026-08-02 | 3500.00      |
| 203      | 1           | 2026-08-03 | 1500.00      |
| 204      | 3           | 2026-08-04 | 5000.00      |
+-----+
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO accounts VALUES(101,'Rahul',50000),(102,'Priya',30000),(103,'Arjun',45000);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 101 | Rahul | 50000.00 |
| 102 | Priya | 30000.00 |
| 103 | Arjun | 45000.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO faculty VALUES(1,'Dr. Reddy'),(2,'Dr. Kumar');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM faculty;
+-----+-----+
| faculty_id | faculty_name |
+-----+-----+
| 1 | Dr. Reddy |
| 2 | Dr. Kumar |
+-----+-----+
2 rows in set (0.00 sec)

mysql> INSERT INTO courses VALUES(1,'DBMS',1),(2,'Java',2);
Query OK, 2 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM courses;
+-----+-----+-----+
| course_id | course_name | faculty_id |
+-----+-----+-----+
| 1 | DBMS | 1 |
| 2 | Java | 2 |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> INSERT INTO students VALUES(1,'Anil'),(2,'Sunil');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM students;
+-----+-----+
| student_id | student_name |
+-----+-----+
| 1 | Anil |
| 2 | Sunil |
+-----+-----+
2 rows in set (0.00 sec)

mysql> INSERT INTO enrollments VALUES(1,1,1),(2,2,2);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO enrollments VALUES(1,1,1),(2,2,2);
ERROR 1062 (23000): Duplicate entry '1' for key 'enrollments.PRIMARY'
mysql> SELECT * FROM enrollments;
```

```
+-----+-----+-----+
| enrollment_id | student_id | course_id |
+-----+-----+-----+
| 1 | 1 | 1 |
| 2 | 2 | 2 |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT emp_name,dept_name FROM employees e INNER JOIN departments d ON e.dept_id=d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Amit | CSE |
| Ravi | CSE |
| Neha | ECE |
| Kiran | EEE |
+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email | role_name |
+-----+-----+-----+
| Rahul | rahul@gmail.com | Admin |
| Priya | priya@gmail.com | Manager |
| Arjun | arjun@gmail.com | Employee |
| Sneha | sneha@gmail.com | NULL |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT name,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+
| name | role_name |
+-----+-----+
| Rahul | Admin |
| Priya | Manager |
| Arjun | Employee |
+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id UNION SELECT name,email,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email | role_name |
+-----+-----+-----+
| Rahul | rahul@gmail.com | Admin |
| Priya | priya@gmail.com | Manager |
| Arjun | arjun@gmail.com | Employee |
| Sneha | sneha@gmail.com | NULL |
+-----+-----+-----+
4 rows in set (0.01 sec)

mysql> SELECT e.emp_name Employee,m.emp_name Manager FROM employees e LEFT JOIN employees m ON e.manager_id=m.emp_id;
```

```
+-----+-----+
| Employee | Manager |
+-----+-----+
| Amit | NULL |
| Ravi | Amit |
| Neha | Amit |
| Kiran | Ravi |
+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT o.order_id,customer_name,city,total_amount FROM customers c JOIN orders o ON c.customer_id=o.customer_id;
+-----+-----+-----+-----+
| order_id | customer_name | city | total_amount |
+-----+-----+-----+-----+
| 201 | Ramesh | Hyderabad | 2500.00 |
| 202 | Sita | Vijayawada | 3500.00 |
| 203 | Ramesh | Hyderabad | 1500.00 |
| 204 | Arun | Vizag | 5000.00 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT dept_name,COUNT(*),AVG(salary),MAX(salary) FROM employees e JOIN departments d ON e.dept_id=d.dept_id GROUP BY dept_name;
+-----+-----+-----+-----+
| dept_name | COUNT(*) | AVG(salary) | MAX(salary) |
+-----+-----+-----+-----+
| CSE | 2 | 70000.000000 | 80000.00 |
| ECE | 1 | 45000.000000 | 45000.00 |
| EEE | 1 | 55000.000000 | 55000.00 |
+-----+-----+-----+-----+
```

```
3 rows in set (0.01 sec)

mysql> SELECT emp_name,salary,dept_name FROM employees e JOIN departments d ON e.dept_id=d.dept_id WHERE salary>50000 AND dept_name='CSE';
+-----+-----+-----+
| emp_name | salary | dept_name |
+-----+-----+-----+
| Amit | 80000.00 | CSE |
| Ravi | 60000.00 | CSE |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT customer_name,SUM(total_amount) AS TotalPurchase FROM customers c JOIN orders o ON c.customer_id=o.customer_id GROUP BY customer_name ORDER BY TotalPurchase DESC;
+-----+-----+
| customer_name | TotalPurchase |
+-----+-----+
| Arun | 5000.00 |
| Ramesh | 4000.00 |
| Sita | 3500.00 |
+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT student_name,course_name,faculty_name FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
```

```

+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Anil         | DBMS        | Dr. Reddy    |
| Sunil        | Java        | Dr. Kumar    |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> START TRANSACTION; INSERT INTO accounts VALUES(104,'Anil',25000); COMMIT;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> START TRANSACTION; INSERT INTO accounts VALUES(105,'Ravi',10000); ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 101        | Rahul       | 50000.00 |
| 102        | Priya       | 30000.00 |
| 103        | Arjun       | 45000.00 |
| 104        | Anil        | 25000.00 |
+-----+-----+-----+

```

```

mysql> START TRANSACTION; UPDATE accounts SET balance=balance-5000 WHERE account_no=101; UPDATE accounts SET balance=balance+5000 WHERE account_no=102; COMMIT;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 101        | Rahul       | 45000.00 |
| 102        | Priya       | 35000.00 |
| 103        | Arjun       | 45000.00 |
| 104        | Anil        | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> START TRANSACTION; UPDATE accounts SET balance=balance-3000 WHERE account_no=101; UPDATE accounts SET balance=balance+3000 WHERE account_no=999; ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

```

```

4 rows in set (0.00 sec)

```

```

mysql> START TRANSACTION; UPDATE accounts SET balance=balance+2000 WHERE account_no=101; SAVEPOINT s1; UPDATE accounts SET balance=balance-1000 WHERE account_no=101; ROLLBACK TO s1; COMMIT;
Query OK, 0 rows affected (0.00 sec)

```

```

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

Query OK, 0 rows affected (0.00 sec)

```

```

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

Query OK, 0 rows affected (0.00 sec)

```

```

Query OK, 0 rows affected (0.01 sec)

```

```

mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 101        | Rahul       | 47000.00 |
| 102        | Priya       | 35000.00 |
| 103        | Arjun       | 45000.00 |
| 104        | Anil        | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)

```

2520030583

Join and Transactions 2

```
Microsoft Windows [Version 10.0.26200.8973]
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C:\Users\DELL>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases
-> ^C
mysql> show databases;
+-----+
| Database |
+-----+
| bookflow_week2 |
| information_schema |
| jt_pt1 |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.38 sec)

mysql> create database DBMS_Transactions2;
Query OK, 1 row affected (0.07 sec)

mysql> use DBMS_Transactions;
ERROR 1049 (42000): Unknown database 'dbms_transactions'
mysql> use DBMS_Transactions2;
Database changed
mysql> create table departments(dept_id int primary key,dept_name varchar(50));
Query OK, 0 rows affected (0.14 sec)
```

```
mysql> create table employees(emp_id int primary key, emp_name varchar(100),salary decimal(10,2),dept_id int ,manager_id int, foreign key (dept_id)reference
s department employees(emp_id));
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'e
mployees(emp_id))' at line 1
mysql> create table employees(emp_id int primary key, emp_name varchar(50),email varchar(100),salary decimal(10,2),dept_id int ,manager_id int, foreign key
(dept_id)references department employees(emp_id));
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'e
mployees(emp_id))' at line 1
mysql> create table employees(emp_id int primary key, emp_name varchar(50),email varchar(100),salary decimal(10,2),dept_id int ,manager_id int, foreign key
(dept_id)references departments(dept_id),foreign key(manager_id) references employees(emp_id));
Query OK, 0 rows affected (0.21 sec)

mysql> create table projects(project_id int primary key,project_name varchar(100),dept_id int,foreign key(dept_id) references departments(dept_id));
Query OK, 0 rows affected (0.12 sec)

mysql> create table customers(customer_id int primary key,customer_name varchar(50),email varchar(100));
Query OK, 0 rows affected (0.06 sec)

mysql> create table products( product_id int primary key,product_name varchar(100),price decimal(10,2));
Query OK, 0 rows affected (0.04 sec)

mysql> create table orders(order_id int primary key,customer_id int,order_date date, foreign key(customer_id) references customers(customer_id));
Query OK, 0 rows affected (0.11 sec)

mysql> create table order_details( order_detail_id int primary key,order_id int,product_id int,quantity int,foreign key (order_id) references orders(order_i
d), foreign key (product_id) references products(product_id));
ERROR 1024 (HY000): Failed to open the referenced table 'product'
mysql> create table order_details( order_detail_id int primary key,order_id int,product_id int,quantity int,foreign key (order_id) references orders(order_i
d), foreign key (product_id) references products(product_id));
Query OK, 0 rows affected (0.13 sec)
```

```
mysql> insert into departments values(101,'cse'),(102,'ece'),(103,'mech'),(104,'civil');
Query OK, 4 rows affected (0.05 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Employees VALUES (1,'Rahul','rahul@gmail.com',70000,101,NULL),(2,'Priya','priya@gmail.com',50000,101,1),(3,'Kiran','kiran@gmail.com',45000,101,1),(4,'Amit','amit@gmail.com',65000,101,1),(5,'Rohit','rohit@gmail.com',75000,101,1),(6,'Sneha','sneha@gmail.com',80000,NULL,NULL);
Query OK, 6 rows affected (0.01 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Projects VALUES (201,'Library Management',101),(202,'Hospital System',102),(203,'Online Shopping',103);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Customers VALUES (1,'Arun','arun@gmail.com'),(2,'Meena','meena@gmail.com'),(3,'David','david@gmail.com'),(4,'Riya','riya@gmail.com');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Products VALUES (101,'Laptop',60000),(102,'Mouse',700),(103,'Keyboard',1200);
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Orders VALUES (1001,1,'2025-01-10'),(1002,2,'2025-01-15'),(1003,1,'2025-02-05'),(1004,3,'2025-02-20');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Order_Details VALUES (1,1001,101,1),(2,1001,102,2),(3,1002,102,5),(4,1003,103,3),(5,1004,102,4);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Order_Details VALUES (1,1001,101,1),(2,1001,102,2),(3,1002,102,5),(4,1003,103,3),(5,1004,102,4);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
mysql> INSERT INTO Employees VALUES (7,'Anjali','anjali@gmail.com',55000,101,1);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Employees
-> VALUES (8,'Vikram','vikram@gmail.com',55000,102,1);
Query OK, 1 row affected (0.01 sec)

mysql> select * from employees;
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email | salary | dept_id | manager_id |
+-----+-----+-----+-----+-----+-----+
| 1 | Rahul | rahul@gmail.com | 70000.00 | 101 | NULL |
| 2 | Priya | priya@gmail.com | 50000.00 | 101 | 1 |
| 3 | Kiran | kiran@gmail.com | 45000.00 | 101 | 1 |
| 4 | Amit | amit@gmail.com | 65000.00 | 101 | 1 |
| 5 | Rohit | rohit@gmail.com | 75000.00 | 101 | 1 |
| 6 | Sneha | sneha@gmail.com | 80000.00 | NULL | NULL |
| 7 | Anjali | anjali@gmail.com | 55000.00 | 101 | 1 |
| 8 | Vikram | vikram@gmail.com | 55000.00 | 102 | 1 |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.01 sec)

mysql> update employees set salary=salary*1.10 where dept_id=101;
Query OK, 6 rows affected (0.01 sec)
Rows matched: 6 Changed: 6 Warnings: 0

mysql> selsect * from employees where dept_id=101;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'selsect * from employees where dept_id=101' at line 1
mysql> select * from employees where dept_id=101;
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email | salary | dept_id | manager_id |
+-----+-----+-----+-----+-----+-----+
| 1 | Rahul | rahul@gmail.com | 77000.00 | 101 | NULL |
| 2 | Priya | priya@gmail.com | 55000.00 | 101 | 1 |
| 3 | Kiran | kiran@gmail.com | 49500.00 | 101 | 1 |
| 4 | Amit | amit@gmail.com | 71500.00 | 101 | 1 |
| 5 | Rohit | rohit@gmail.com | 82500.00 | 101 | 1 |
| 7 | Anjali | anjali@gmail.com | 60500.00 | 101 | 1 |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> delete from employees where salary<30000;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from employees;
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email | salary | dept_id | manager_id |
+-----+-----+-----+-----+-----+-----+
| 1 | Rahul | rahul@gmail.com | 77000.00 | 101 | NULL |
| 2 | Priya | priya@gmail.com | 55000.00 | 101 | 1 |
| 3 | Kiran | kiran@gmail.com | 49500.00 | 101 | 1 |
| 4 | Amit | amit@gmail.com | 71500.00 | 101 | 1 |
| 5 | Rohit | rohit@gmail.com | 82500.00 | 101 | 1 |
| 6 | Sneha | sneha@gmail.com | 80000.00 | NULL | NULL |
| 7 | Anjali | anjali@gmail.com | 60500.00 | 101 | 1 |
| 8 | Vikram | vikram@gmail.com | 55000.00 | 102 | 1 |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> select e.emp_name ,d.dept_name from employees e inner join departments d on e.dept_id=d.dept_id;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'select e.emp_name ,d.dept_name from employees e inner join departments d on e.de' at line 1
mysql> select e.emp_name ,d.dept_name from employees e inner join departments d on e.dept_id=d.dept_id;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'select e.emp_name ,d.dept_name from employees e inner join departments d on e.d' at line 1
mysql> select e.emp_name ,d.dept_name from employees e inner join departments d on e.dept_id=d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Rahul | cse |
| Priya | cse |
| Kiran | cse |
| Amit | cse |
| Rohit | cse |
| Anjali | cse |
| Vikram | ece |
+-----+-----+
7 rows in set (0.00 sec)

mysql> select e.emp_name,d.dept_name from employees e left join departments d on e.dept_id = d.dept_id where d.dept_id is Null;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Sneha | NULL |
+-----+-----+
1 rows in set (0.00 sec)
```

```

1 row in set (0.01 sec)

mysql> nxt
-> ^C
mysql> select d.dept_name,e.emp_name from employees e right join departments d on e.dept_id =d.dept_id;
+-----+-----+
| dept_name | emp_name |
+-----+-----+
| cse       | Rahul    |
| cse       | Priya    |
| cse       | Kiran    |
| cse       | Amit     |
| cse       | Rohit    |
| cse       | Anjali   |
| ece       | Vikram   |
| mech      | NULL     |
| civil     | NULL     |
+-----+-----+
9 rows in set (0.01 sec)

mysql> SELECT E.Emp_Name AS Employee, M.Emp_Name AS Manager
-> FROM Employees E
-> LEFT JOIN Employees M
-> ON E.Manager_ID = M.Emp_ID;
+-----+-----+
| Employee | Manager |
+-----+-----+
| Rahul    | NULL    |
| Priya    | Rahul   |
| Kiran    | Rahul   |
| Amit     | Rahul   |
| Rohit    | Rahul   |
| Sneha    | NULL    |
| Anjali   | Rahul   |
| Vikram   | Rahul   |
+-----+-----+
8 rows in set (0.00 sec)

```

```

mysql> SELECT COUNT(*) AS Total_Employees
-> select d.dept_name,e.emp_name from employees e right join departments d on e.dept_id =d.dept_id;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'select d.dept_n
ame,e.emp_name from employees e right join departments d on e.dept_id =d.dept_id' at line 2
mysql> select count(*) as total_employees from employees;
+-----+
| total_employees |
+-----+
| 8               |
+-----+
1 row in set (0.01 sec)

mysql> select dept_id, avg(salary) as avg_salary from employees group by dept_id;
+-----+-----+
| dept_id | avg_salary |
+-----+-----+
| NULL    | 80000.000000 |
| 101     | 66000.000000 |
| 102     | 55000.000000 |
+-----+-----+
3 rows in set (0.01 sec)

mysql> select dept_id, avg(salary) as avg_salary from employees group by dept_id having avg(salary) > 60000;
+-----+-----+
| dept_id | avg_salary |
+-----+-----+
| NULL    | 80000.000000 |
| 101     | 66000.000000 |
+-----+-----+
2 rows in set (0.01 sec)

mysql> select max(salary) as highest_salary, min(salary) as lowest_salary from employees;
+-----+-----+
| highest_salary | lowest_salary |
+-----+-----+
| 82500.00       | 49500.00      |
+-----+-----+
1 row in set (0.01 sec)

```

```
mysql> select * from employees where salary > (select avg(salary) from employees);
```

emp_id	emp_name	email	salary	dept_id	manager_id
1	Rahul	rahul@gmail.com	77000.00	101	NULL
4	Amit	amit@gmail.com	71500.00	101	1
5	Rohit	rohit@gmail.com	82500.00	101	1
6	Sneha	sneha@gmail.com	80000.00	NULL	NULL

```
4 rows in set (0.01 sec)
```

```
mysql> select * from employees where dept_id=(select dept_id from employees where emp_name='rahul');
```

emp_id	emp_name	email	salary	dept_id	manager_id
1	Rahul	rahul@gmail.com	77000.00	101	NULL
2	Priya	priya@gmail.com	55000.00	101	1
3	Kiran	kiran@gmail.com	49500.00	101	1
4	Amit	amit@gmail.com	71500.00	101	1
5	Rohit	rohit@gmail.com	82500.00	101	1
7	Anjali	anjali@gmail.com	60500.00	101	1

```
6 rows in set (0.01 sec)
```

```
mysql> select * from employees where salary=(select max(salary) from employees);
```

emp_id	emp_name	email	salary	dept_id	manager_id
5	Rohit	rohit@gmail.com	82500.00	101	1

```
1 row in set (0.01 sec)
```

```
mysql> select dept_id,count(*) as total_employees from employees group by dept_id having count(*)>5;
```

dept_id	total_employees
101	6

```
1 row in set (0.00 sec)
```

```
mysql> select distinct salary from (select salary,dense_rank() over(order by salary desc) as rnk from employees) t where rnk=2;
```

salary
80000.00

```
1 row in set (0.02 sec)
```

```
mysql> select * from employees e where salary>(select avg(salary) from employees where dept_id=e.dept_id);
```

emp_id	emp_name	email	salary	dept_id	manager_id
1	Rahul	rahul@gmail.com	77000.00	101	NULL
4	Amit	amit@gmail.com	71500.00	101	1
5	Rohit	rohit@gmail.com	82500.00	101	1

```
3 rows in set (0.01 sec)
```

```
mysql> select * from customers where customer_id not in(select customer_id from orders);
```

customer_id	customer_name	email
4	Riya	riya@gmail.com

```
1 row in set (0.01 sec)
```

```
mysql> select p.product_name,sum(od.quantity) as total_sold from products p join order_details od on p.product_id=od.product_id group by p.product_name order by total_sold desc limit 1;
```

product_name	total_sold
Mouse	11

```
1 row in set (0.01 sec)
```

```
mysql> select dept_id,count(*) as total_employees from employees group by dept_id order by total_employees desc limit 1;
```

dept_id	total_employees
101	6

```
1 row in set (0.01 sec)
```

```
mysql> select month(order_date) as month, sum(quantity) over(partition by month(order_date)) as monthly_sales from orders o join order_details od on o.order_id=od.order_id
;
+-----+-----+
| month | monthly_sales |
+-----+-----+
| 1     | 8             |
| 1     | 8             |
| 1     | 8             |
| 2     | 7             |
| 2     | 7             |
+-----+-----+
5 rows in set (0.01 sec)
```

```
mysql> select c.customer_name, sum(p.price*od.quantity) as total_purchase, rank() over(order by sum(p.price*od.quantity) desc) as rnk from customers c join orders o on c.customer_id=o.customer_id join order_details od on o.order_id=od.order_id join products p on od.product_id=p.product_id group by c.customer_name;
+-----+-----+-----+
| customer_name | total_purchase | rnk |
+-----+-----+-----+
| Arun          | 65000.00      | 1   |
| Meena         | 3500.00       | 2   |
| David         | 2800.00       | 3   |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> select e.emp_name, m.emp_name as manager from employees e join employees m on e.manager_id=m.emp_id where e.salary>m.salary;
+-----+-----+
| emp_name | manager |
+-----+-----+
| Rohit    | Rahul   |
+-----+-----+
1 row in set (0.01 sec)
```

```
mysql> select email, count(*) from employees group by email having count(*)>1;
Empty set (0.00 sec)
```

```
mysql> select email, count(*) from employees group by email having count(*)>1;
Empty set (0.00 sec)
```

```
mysql> select * from departments;
```

dept_id	dept_name
101	cse
102	ece
103	mech
104	civil

4 rows in set (0.01 sec)

```
mysql> select * from employees;
```

emp_id	emp_name	email	salary	dept_id	manager_id
1	Rahul	rahul@gmail.com	77000.00	101	NULL
2	Priya	priya@gmail.com	55000.00	101	1
3	Kiran	kiran@gmail.com	49500.00	101	1
4	Amit	amit@gmail.com	71500.00	101	1
5	Rohit	rohit@gmail.com	82500.00	101	1
6	Sneha	sneha@gmail.com	80000.00	NULL	NULL
7	Anjali	anjali@gmail.com	60500.00	101	1
8	Vikram	vikram@gmail.com	55000.00	102	1

8 rows in set (0.00 sec)

```
mysql> select * from projects;
```

project_id	project_name	dept_id
201	Library Management	101
202	Hospital System	102
203	Online Shopping	103

3 rows in set (0.01 sec)

```
mysql> select * from customers;
```

customer_id	customer_name	email
1	Arun	arun@gmail.com
2	Meena	meena@gmail.com
3	David	david@gmail.com
4	Riya	riya@gmail.com

4 rows in set (0.00 sec)

4 rows in set (0.00 sec)

```
mysql> select * from products;
```

product_id	product_name	price
101	Laptop	60000.00
102	Mouse	700.00
103	Keyboard	1200.00

3 rows in set (0.01 sec)

```
mysql> select * from orders;
```

order_id	customer_id	order_date
1001	1	2025-01-10
1002	2	2025-01-15
1003	1	2025-02-05
1004	3	2025-02-20

4 rows in set (0.00 sec)

```
mysql> select * from order_details;
```

order_detail_id	order_id	product_id	quantity
1	1001	101	1
2	1001	102	2
3	1002	102	5
4	1003	103	3
5	1004	102	4

5 rows in set (0.01 sec)

```
mysql> |
```